AGO2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes AGO2 gene fusions from the msk_impact_50k_2026 study, covering 23 fusion events. 6/13 fusions with known frame status (46.2%) were in-frame. 7/22 fusions (31.8%) retained the Piwi domain. Domain retention was tested with Fisher's exact test (p=0.225232, not statistically significant at alpha=0.05). No literature benchmark is configured for AGO2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 23 analyzed fusion events, identifying 8 distinct fusion partner genes. The most frequent partner was TRAPPC9, observed in 12 events.

Domain retention

Piwi domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/6 (50.0%) of in-frame fusions retained the domain, p=0.225232 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.841584.

Domain disruption

Disruption of the Argonaute linker 1 domain and PAZ domain was tested with Fisher's exact test comparing in-frame fusions against all others; 4/6 (66.7%) of in-frame fusions disrupted the domain, p=0.314241 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.128713.

Junction concentration

22% of 9 fusions with this gene as the 3' partner join at exon 9, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.034965, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 20 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 468 aa (exon 11/12 boundary; label-permutation p=0.0559441). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -50 aa from the nearest configured domain boundary at 518 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=6) and "not_retained" (n=14). For Frame_status, retained: 3/5 (60.0%, 95% CI 23.1%-88.2%); not_retained: 3/7 (42.9%, 95% CI 15.8%-75.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 27.3333 vs 18.0714, p=0.394505 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for AGO2 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (13 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: AGO2 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: AGO2 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (41 row(s) in window_scan, 28 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRAPPC9	12	0.5217391304347826	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2518270876777048	1
PTK2	5	0.21739130434782608	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.14441020788230835	2
COL22A1	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	3
FER1L6	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	4
MRPL48	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	5
POLB	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	6
SAMD12	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	7
SLX4	1	0.043478260869565216	0.007490244806834985	0.08903780214758059	None	0.37851163764389395	confidence_certainty, domain_disruption, domain_retention, recurrence	0.08302913371351038	8

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
8	0	6	1	13	0.0	1.0	-0.0
72	1	5	1	13	2.6	0.5210526315789474	0.283118406355279
101	1	5	2	12	1.2	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	2	4	2	12	3.0	0.5492260061919505	0.26024890683539526
227	2	4	3	11	1.8333333333333333	0.6126160990712074	0.2128115944606906
264	2	4	4	10	1.25	1.0	-0.0
343	4	2	4	10	5.0	0.16109391124871003	0.7929208739883478
423	4	2	5	9	3.6	0.33591331269349844	0.4737727841465546
435	5	1	5	9	9.0	0.14086687306501547	0.8511911256739905
468	6	0	5	9	None	0.014086687306501547	1.8511911256739906
483	6	0	6	8	None	0.04189886480908153	1.377797743473571
509	6	0	7	7	None	0.05149638802889577	1.2882323314273754
578	6	0	8	6	None	0.11488648090815273	0.9397310732475048
583	6	0	9	5	None	0.26044891640866874	0.5842774450990696
613	6	0	10	4	None	0.2674922600619195	0.5726887798522096
653	6	0	11	3	None	0.5210526315789474	0.283118406355279
679	6	0	12	2	None	1.0	-0.0
723	6	0	13	1	None	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

4	6
2	8

frame_domain_contingency_tables_by_tumor_type

stratum	table
ASTR	[0, 0], [0, 1]
BLCA	[0, 1], [0, 0]
BRCA	[0, 1], [0, 1]
BRCNOS	[1, 0], [0, 0]
CUP	[1, 0], [0, 0]
ESCA	[0, 0], [1, 0]
HGSOC	[1, 0], [0, 2]
IDC	[0, 2], [0, 2]
LMS	[0, 1], [0, 0]
LUAD	[1, 0], [0, 0]

stratum	table
OS	[0, 0], [0, 1]
UCS	[0, 1], [0, 0]
UEC	[0, 0], [0, 1]
ULMS	[0, 0], [1, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf08699	Argonaute linker 1 domain	20	10	1	9	0.016981132075471698
auto_disruption_pf02170	PAZ domain	20	9	3	8	0.05748663101604278

domain_retention tables

frame_domain_contingency_table

3	3
3	11

frame_domain_contingency_tables_by_tumor_type

stratum	table
ASTR	[0, 1], [0, 0]
BLCA	[0, 0], [0, 1]
BRCA	[0, 0], [0, 2]
BRCNOS	[1, 0], [0, 0]
CUP	[0, 0], [1, 0]
ESCA	[0, 0], [1, 0]
HGSOC	[1, 0], [0, 2]
IDC	[0, 1], [0, 3]
LMS	[0, 0], [0, 1]
LUAD	[1, 0], [0, 0]
OS	[0, 0], [0, 1]
UCS	[0, 1], [0, 0]
UEC	[0, 0], [0, 1]
ULMS	[0, 0], [1, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Piwi domain	20	6	7	7	0.18880952380952382

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	10	21	0.47619047619047616

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
COL22A1	1	1	1	1
FER1L6	1	1	1	1
MRPL48	1	1	1	1
POLB	1	1	1	1
PTK2	5	5	5	5
SAMD12	1	1	1	1
SLX4	1	1	1	1
TRAPPC9	12	12	12	10

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
8	141559000	141560000	3	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0061336-T01-IM7-45
8	141549000	141550000	2	EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42
8	141557000	141558000	2	EVT-P-0024944-T01-IM6-40, EVT-P-0066481-T01-IM7-33
8	141561000	141562000	2	EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31
8	141566000	141567000	2	EVT-P-0039584-T01-IM6-35, EVT-P-0059625-T01-IM7-43

chromosome	bin_start	bin_end	n_events	event_ids
8	141569000	141570000	2	EVT-P-0005188-T02-IM6-25, EVT-P-0066230-T02-IM7-32
8	141542000	141543000	1	EVT-P-0019967-T01-IM6-47
8	141551000	141552000	1	EVT-P-0016571-T01-IM6-26
8	141554000	141555000	1	EVT-P-0037757-T01-IM6-24
8	141568000	141569000	1	EVT-P-0063858-T02-IM7-30
8	141582000	141583000	1	EVT-P-0015814-T05-IM7-27
8	141595000	141596000	1	EVT-P-0065414-T01-IM7-36
8	141645000	141646000	1	EVT-P-0066796-T02-IM7-28

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
343	2	2	151	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	6	2	11	0.18181818181818182	0.009161840026805802	2.359544712253732
five_prime	12	2	11	0.18181818181818182	0.015997319419845866	1.8932238274470852
five_prime	13	2	11	0.18181818181818182	0.03182231582978316	1.3370606283467543
three_prime	9	2	9	0.2222222222222222	0.008826767507539132	2.569996220280189
five_prime	1	1	11	0.09090909090909091	0.4816236656933608	0.0003155287194407593
five_prime	3	1	11	0.09090909090909091	0.09858790866880475	0.16702466861450302
five_prime	10	1	11	0.09090909090909091	0.04386577952228232	0.4095267320662918
five_prime	14	1	11	0.09090909090909091	0.028193959121152648	0.5687786933813146
five_prime	16	1	11	0.09090909090909091	0.03719304963860035	0.4673344729247356
three_prime	3	1	9	0.1111111111111111	0.1178210712747116	0.16979717091013546
three_prime	6	1	9	0.1111111111111111	0.009372457038916279	1.0901340770938877
three_prime	11	1	9	0.1111111111111111	0.04397108802833756	0.4777830015744832
three_prime	12	1	9	0.1111111111111111	0.02095639270499258	0.760583668745073
three_prime	15	1	9	0.1111111111111111	0.029189603178402183	0.6307219483206221
three_prime	16	1	9	0.1111111111111111	0.017653535015078263	0.8294079089709719
three_prime	18	1	9	0.1111111111111111	0.02921832368005361	0.6303437012174128

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
27	227	200	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0065414-T01-IM7-36, EVT-P-0066230-T02-IM7-32
64	264	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0063858-T02-IM7-30, EVT-P-0065414-T01-IM7-36, EVT-P-0066230-T02-IM7-32
72	272	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0063858-T02-IM7-30, EVT-P-0065414-T01-IM7-36, EVT-P-0066230-T02-IM7-32
101	301	200	4	16	1	5	0.7333333333333333	1.0	-0.0	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
143	343	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0005188-T02-IM6-25, EVT-P-0039584-T01-IM6-35, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
219	419	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0005188-T02-IM6-25, EVT-P-0039584-T01-IM6-35, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
223	423	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0039584-T01-IM6-35, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
227	427	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0039584-T01-IM6-35, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
235	435	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30
264	464	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
268	468	200	5	15	4	2	26.0	0.01393188854489164	1.8559900085557592	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0061336-T01-IM7-45
283	483	200	6	14	4	2	12.0	0.03740970072239422	1.4270157661437528	EVT-P-0021826-T01-IM6-46, EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, ... (6 total)
309	509	200	7	13	4	2	7.333333333333333	0.12192982456140351	0.9138900510823775	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (7 total)
335	435	100	4	16	3	3	13.0	0.0608875128998968	1.2154717654086211	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43
343	443	100	4	16	3	3	13.0	0.0608875128998968	1.2154717654086211	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43
343	543	200	7	13	4	2	7.333333333333333	0.12192982456140351	0.9138900510823775	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (7 total)
378	578	200	6	14	2	4	1.25	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (6 total)
383	483	100	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0021826-T01-IM6-46, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0061336-T01-IM7-45
383	583	200	7	13	2	4	0.9	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (7 total)
409	509	100	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0061336-T01-IM7-45

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
413	613	200	8	12	2	4	0.6666666666666666	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0059084-T01-IM7-41, ... (8 total)
423	523	100	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0061336-T01-IM7-45
423	623	200	8	12	2	4	0.6666666666666666	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0059084-T01-IM7-41, ... (8 total)
435	535	100	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0061336-T01-IM7-45
435	635	200	7	13	2	4	0.9	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0059084-T01-IM7-41, ... (7 total)
453	653	200	7	13	1	5	0.26666666666666666	0.354360165118679	0.45055510462214704	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)
468	668	200	7	13	1	5	0.26666666666666666	0.354360165118679	0.45055510462214704	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)
479	679	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)
483	583	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0066481-T01-IM7-33
483	683	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
509	709	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0063424-T01-IM7-42, ... (6 total)
523	723	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42, ... (6 total)
553	653	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0066481-T01-IM7-33
558	758	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, ... (7 total)
578	678	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0066481-T01-IM7-33
578	778	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, ... (7 total)
579	679	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0037757-T01-IM6-24, EVT-P-0063424-T01-IM7-42, EVT-P-0066481-T01-IM7-33
583	683	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0037757-T01-IM6-24, EVT-P-0063424-T01-IM7-42, EVT-P-0066481-T01-IM7-33
583	783	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42, ... (6 total)
613	813	200	5	15	0	6	0.0	0.26044891640866874	0.5842774450990696	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
653	853	200	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
268	468	200	5	15	4	2	26.0	0.01393188854489164	1.8559900085557592	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0061336-T01-IM7-45
283	483	200	6	14	4	2	12.0	0.03740970072239422	1.4270157661437528	EVT-P-0021826-T01-IM6-46, EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, ... (6 total)
479	679	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)
558	758	200	7	13	0	6	0.0	0.051496388028895765	1.2882232314273754	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, ... (7 total)
335	435	100	4	16	3	3	13.0	0.0608875128998968	1.2154717654086211	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43
509	709	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0063424-T01-IM7-42, ... (6 total)
523	723	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42, ... (6 total)
583	783	200	6	14	0	6	0.0	0.11488648090815273	0.9397310732475048	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
309	509	200	7	13	4	2	7.333333333333333	0.12192982456140351	0.9138900510823775	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (7 total)
143	343	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0005188-T02-IM6-25, EVT-P-0039584-T01-IM6-35, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
235	435	200	5	15	3	3	6.0	0.13132094943240455	0.881665986041969	EVT-P-0039584-T01-IM6-35, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30
613	813	200	5	15	0	6	0.0	0.26044891640866874	0.5842774450990696	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0037757-T01-IM6-24, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42
483	583	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0066481-T01-IM7-33
553	653	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0024944-T01-IM6-40, EVT-P-0037757-T01-IM6-24, EVT-P-0066481-T01-IM7-33
579	679	100	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0037757-T01-IM6-24, EVT-P-0063424-T01-IM7-42, EVT-P-0066481-T01-IM7-33
653	853	200	4	16	0	6	0.0	0.2674922600619195	0.5726887798522096	EVT-P-0016571-T01-IM6-26, EVT-P-0019967-T01-IM6-47, EVT-P-0039159-T01-IM6-29, EVT-P-0063424-T01-IM7-42
453	653	200	7	13	1	5	0.26666666666666666	0.354360165118679	0.45055510462214704	EVT-P-0016571-T01-IM6-26, EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, ... (7 total)
383	483	100	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0021826-T01-IM6-46, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0061336-T01-IM7-45
435	535	100	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0061336-T01-IM7-45

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
27	227	200	4	16	2	4	3.0	0.5492260061919505	0.26024890683539526	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0065414-T01-IM7-36, EVT-P-0066230-T02-IM7-32
409	509	100	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0021826-T01-IM6-46, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, EVT-P-0061336-T01-IM7-45
64	264	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0063858-T02-IM7-30, EVT-P-0065414-T01-IM7-36, EVT-P-0066230-T02-IM7-32
223	423	200	5	15	2	4	1.8333333333333333	0.6126160990712074	0.2128115944606906	EVT-P-0039584-T01-IM6-35, EVT-P-0059476-T01-IM7-31, EVT-P-0059625-T01-IM7-43, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
101	301	200	4	16	1	5	0.7333333333333333	1.0	-0.0	EVT-P-0005188-T02-IM6-25, EVT-P-0015814-T05-IM7-27, EVT-P-0063858-T02-IM7-30, EVT-P-0066230-T02-IM7-32
378	578	200	6	14	2	4	1.25	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (6 total)
383	583	200	7	13	2	4	0.9	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0059084-T01-IM7-41, EVT-P-0059476-T01-IM7-31, ... (7 total)
413	613	200	8	12	2	4	0.6666666666666666	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0059084-T01-IM7-41, ... (8 total)
435	635	200	7	13	2	4	0.9	1.0	-0.0	EVT-P-0021826-T01-IM6-46, EVT-P-0024944-T01-IM6-40, EVT-P-0035619-T01-IM6-34, EVT-P-0037757-T01-IM6-24, EVT-P-0059084-T01-IM7-41, ... (7 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003775 7-T01-IM6-24	P-003775 7-T01-IM6	P-003775 7	Bone Cancer	OS	AGO2-POLB	POLB	out-of-frame	five_prime	141554151	14	613	True	613	True	exact match	disrupted	0.32	True	partner_lacks_domain	PF02171					40			False		Argonaute linker 1 domain; PAZ domain		Piwi domain	AGO2(NM_012154) - POLB(NM_002690): Rearrangement: c.1839+161: AGO2_c.120-827:POLBinv | Protein Fusion: out of frame{AGO2:POLB}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000518 8-T02-IM6-25	P-000518 8-T02-IM6	P-000518 8	Breast Cancer	BRCNOS	AGO2-PTK2	PTK2	in-frame	three_prime	1415696 96	6	219	True	219	True	exact match	retained	1.0	False													Piwi domain; PAZ domain		Argonaute linker 1 domain	PTK2 (NM_005607) - AGO2 (NM_012154) rearrangement: c.35-42266:PTK2_c.656-68:AGO2del | Protein Fusion: in frame {PTK2:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001657 1-T01-IM6-26	P-001657 1-T01-IM6	P-001657 1	Breast Cancer	IDC	AGO2-PTK2	PTK2	unknown	three_prime	141551339	15	653	False	653	True	exact match	disrupted	0.55	True	partner_lacks_domain	PF02171					677			False			Argonaute linker 1 domain; PAZ domain	Piwi domain	PTK2 (NM_005607) - AGO2 (NM_012154) rearrangement: c.2096+4004:PTK2_c.1958:AGO2del | Protein Fusion: mid-exon {PTK2:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001581 4-T05-IM7-27	P-001581 4	P-001581 4	Soft Tissue Sarcoma	LMS	AGO2-PTK2	PTK2	unknown	five_prime	141582946	3	101	False	100	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Piwi domain; Argonaute linker 1 domain; PAZ domain		AGO2 (NM_012154) - PTK2 (NM_005607) rearrangement: c.301:AGO2_c.35-19:PTK2 dup | Protein Fusion: mid-exon {AGO2:PTK2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006676-T02-IM7-28	P-006676-T02-IM7	P-006676	Bladder Cancer	BLCA	AGO2-PTK2	PTK2	out-of-frame	five_prime	141645211	1	8	True	7	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															Piwi domain; Argonaute linker 1 domain; PAZ domain		AGO2 (NM_012154)-PTK2 (NM_005607) rearrangement: c.22+373:AGO2_c.3012+358:PTK2dup | Protein Fusion: out of frame {AGO2:PTK2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003915 9-T01-IM6-29	P-003915 9-T01-IM6	P-003915 9	Esophagogastric Cancer	ESCA	AGO2-PTK2	PTK2	in-frame	five_prime	141549343	16	723	True	723	True	exact match	disrupted	0.686666666666666	True	partner_lacks_domain	PF02171					715			False		Argonaute linker 1 domain; PAZ domain		Piwi domain	AGO2 (NM_012154) - PTK2 (NM_005607) rearrangement: c.2169+76:AGO2_c.2208+5073:PTK2 dup | Protein Fusion: in frame {AGO2:PTK2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006385 8-T02-IM7-30	P-006385 8-T02-IM7	P-006385 8	Uterine Sarcoma	ULMS	AGO2-SAMD12	SAMD12	in-frame	five_prime	141568892	6	264	True	263	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False														Argonaute linker 1 domain	Piwi domain	PAZ domain	AGO2 (NM_012154) - SAMD12 (NM_207506) rearrangement: c.791-221:AGO2_c.322+25570:SAMD12del | ProteinFusion: in frame {AGO2:SAMD12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005976-T01-IM7-31	P-005976-T01-IM7	P-005976	Ovarian Cancer	HGSOC	AGO2-TRAPPC9	TRAPPC9	out-of-frame	five_prime	141561581	10	423	True	423	True	exact match	lost	0.0	False													Argonaute linker 1 domain; PAZ domain	Piwi domain		AGO2 (NM_012154) - TRAPPC9 (NM_031466) rearrangement: c.1270-46:AGO2_c.1181-7719:TRAPPC9del | Protein Fusion: out of frame {AGO2:TRAPPC9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006623 0-T02-IM7 -32	P-006623 0-T02-IM7	P-006623 0	Endometrial Cancer	UEC	AGO2-TRAPPC9	TRAPPC9	unknown	five_prime	141569605	6	227	False	226	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															Argonaute linker 1 domain	Piwi domain; PAZ domain		AGO2 (NM_012154) - TRAPPC9 (NM_031466) rearrangement: c.679:AGO2_c.1429-1 2206:TRAPPC9del | Protein Fusion: mid-exon { AGO2:TRAPPC9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006681-T01-IM7-33	P-006681-T01-IM7	P-006681	Ovarian Cancer	HGSOC	AGO2-TRAPPC9	TRAPPC9	out-of-frame	five_prime	141557415	13	583	True	582	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.22	True	partner_lacks_domain	PF02171					549			False		Argonaute linker 1 domain; PAZ domain			Piwidomain	AGO2 (NM_012154) - TRAPPC9 (NM_031466) rearrangement: c.1748+152:AGO2_c.1645+3948:TRAPPC9del | Protein Fusion: out of frame {AGO2:TRAPPC9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003561 9-T01-IM6-34	P-003561 9-T01-IM6	P-003561 9	Breast Cancer	IDC	COL22A1-AGO2	COL22A1	unknown	five_prime	141559275	12	509	False	508	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															Argonaute linker 1 domain; PAZ domain	Piwi domain		AGO2 (NM_012154) - COL22A1 (NM_152888) rearrangement: c.1526:AGO2_c.658+7077:COL22A1 del | Protein Fusion: mid-exon { AGO2:COL22A1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003958 4-T01-IM6-35	P-003958 4-T01-IM6	P-003958 4	Non-Small Cell Lung Cancer	LUAD	FER1L6-AGO2	FER1L6	in-frame	three_prime	141566388	9	343	True	343	True	exact match	retained	1.0	False													Piwi domain	Argonaute linker 1 domain	PAZ domain	FER1L6 (NM_001039112) - AGO2 (NM_012154) rearrangement: c.2589+1237:FER1L6_c.1027-3:AGO2inv | Protein Fusion: in frame {FER1L6:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006514-T01-IM7-36	P-006514-T01-IM7	P-006514	Glioma	ASTR	MRPL48-AGO2	MRPL48	out-of-frame	three_prime	141595215	3	72	True	72	True	exact match	retained	1.0	False													Piwi domain; Argonaute linker 1 domain; PAZ domain			MRPL48 (NM_016055) - AGO2 (NM_012154) rearrangement: t(8;11)(q24.3;q13.4)(chr8:g.141595215::chr11:g.73514482) | Protein Fusion: out of frame {MRPL48: AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002494-T01-IM6-40	P-002494-T01-IM6	P-002494	Breast Cancer	BRCA	TRAPPC9-AGO2	TRAPPC9	unknown	five_prime	141557583	13	578	False	577	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.203333333333334	True	partner_lacks_domain	PF02171						342			False		Argonaute linker 1 domain; PAZ domain		Piwi domain	AGO2 (NM_012154) - TRAPPC9 (NM_003166) rearrangement: c.1732:AGO2_c.1024+1832:TRAPPC9del | Protein Fusion: mid-exon { AGO2:TRAPPC9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005908 4-T01-IM7-41	P-005908 4-T01-IM7	P-005908 4	Breast Cancer	IDC	TRAPPC9-AGO2	TRAPPC9	unknown	three_prime	141561500	11	435	False	435	True	exact match	retained	1.0	False													Piwi domain	Argonaute linker 1 domain; PAZ domain		TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.2850+47628:TRAPPC9_c.1305:AGO2dup | Protein Fusion: mid-exon {TRAPPC9:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006342-T01-IM7-42	P-006342-T01-IM7	P-006342	Cancer of Unknown Primary	CUP	TRAPPC9-AGO2	TRAPPC9	in-frame	three_prime	141549644	16	679	True	679	True	exact match	disrupted	0.463333333333333	True	partner_lacks_domain	PF02171					476			False			Argonaute linker 1 domain; PAZ domain	Piwi domain	TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.1428+8162:TRAPPC9_c.2035-91:AGO2 dup | Protein Fusion: in frame {TRAPPC9:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005965-T01-IM7-43	P-005965-T01-IM7	P-005965	Endometrial Cancer	UCS	TRAPPC9-AGO2	TRAPPC9	out-of-frame	three_prime	141566539	9	343	True	343	True	exact match	retained	1.0	False													Piwi domain	Argonaute linker 1 domain	PAZ domain	TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.1180+144:TRAPPC9_c.1027-154:AGO2 dup | Protein Fusion: out of frame {TRAPPC9:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005965-T02-IM7-44	P-005965-T02-IM7	P-005965	Endometrial Cancer	UMEC	TRAPPC9-AGO2	TRAPPC9	out-of-frame	three_prime	141566539	9	343	True	343	True	exact match	retained	1.0	False													Piwi domain	Argonaute linker 1 domain	PAZ domain	TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.1180+144:TRAPPC9_c.1027-154:AGO2 dup | Protein Fusion: out of frame {TRAPPC9:AGO2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006133 6-T01-IM7-45	P-006133 6-T01-IM7	P-006133 6	Ovarian Cancer	HGSOC	TRAPPC9-AGO2	TRAPPC9	in-frame	three_prime	141559516	12	468	True	468	True	exact match	retained	1.0	False												Piwi domain	Argonaute linker 1 domain; PAZ domain			TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.878+4675:TRAPPC9_c.1404-119:AGO2 dup | Protein Fusion: in frame {TRAPPC9:AGO2}	NA

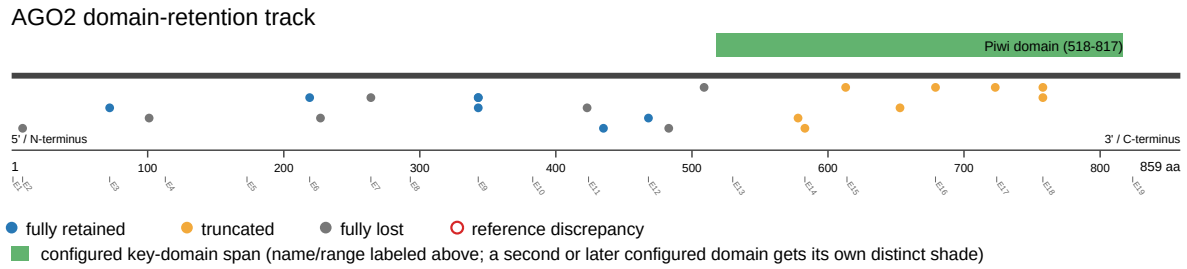
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002182 6-T01-IM6-46	P-002182 6-T01-IM6	P-002182 6	Breast Cancer	IDC	TRAPPC9-AGO2	TRAPPC9	unknown	five_prime	141559353	12	483	False	482	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															Argonaute linker 1 domain; PAZ domain	Piwi domain		AGO2 (NM_012154) - TRAPPC9 (NM_031466) rearrangement: c.1448:AGO2_c.3349+13144:TRAPPC9del | Protein Fusion: mid-exon {AGO2: TRAPPC9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001996 7-T01-IM6-47	P-001996 7-T01-IM6	P-001996 7	Breast Cancer	BRCA	TRAPPC9-AGO2	TRAPPC9	unknown	three_prime	141542713	18	758	False	758	True	exact match	disrupted	0.2	True	partner_lacks_domain	PF02171					1035			False			Argonaute linker 1 domain; PAZ domain	Piwi domain	TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.3105-25933:TRAPPC9_c.2273:AGO2dup | Protein Fusion: mid-exon {TRAPPC9:AGO2}	NA

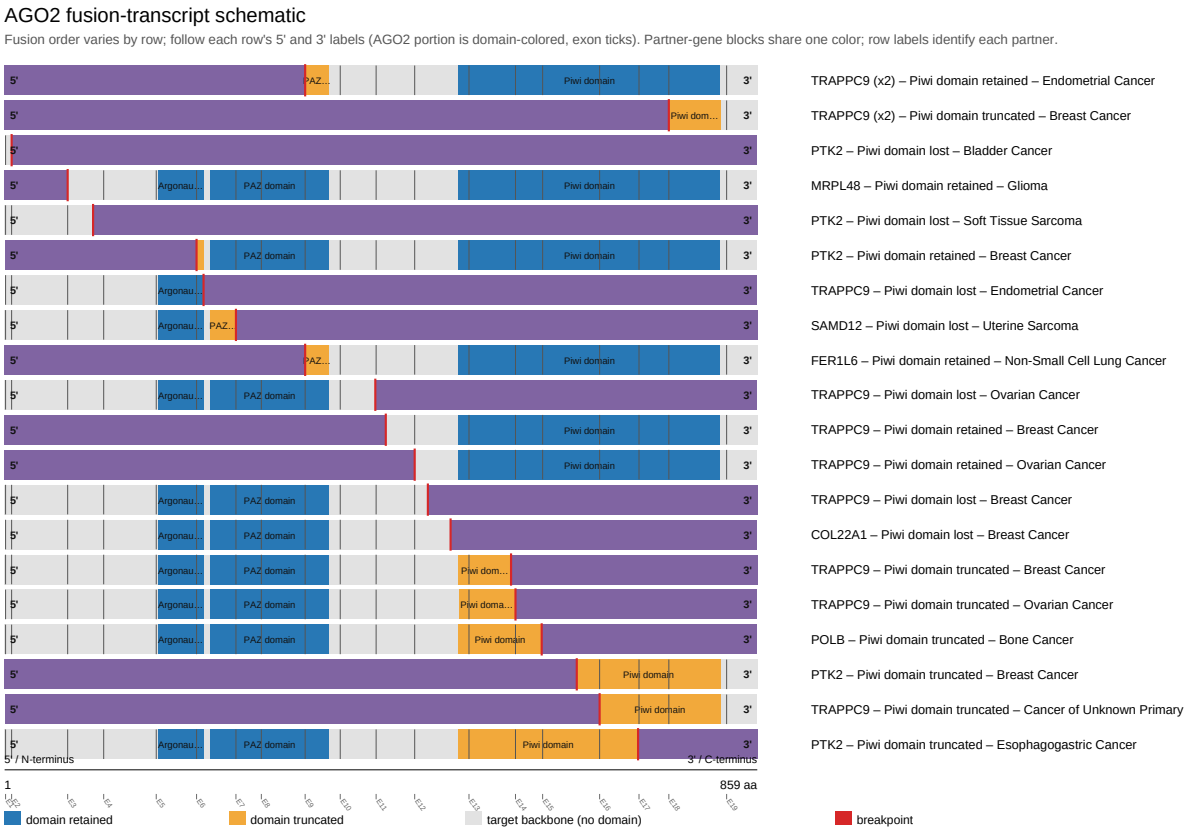
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001996 7-T02-IM6-48	P-001996 7-T02-IM6	P-001996 7	Breast Cancer	IDC	TRAPPC9-AGO2	TRAPPC9	unknown	three_prime	141542713	18	758	False	758	True	exact match	disrupted	0.2	True	partner_lacks_domain	PF02171					1035			False			Argonaute linker 1 domain; PAZ domain	Piwi domain	TRAPPC9 (NM_031466) - AGO2 (NM_012154) rearrangement: c.3105-25933:TRAPPC9_c.2273:AGO2dup | Protein Fusion: mid-exon {TRAPPC9:AGO2}	NA

Figures

domain retention outliers



fusion schematic

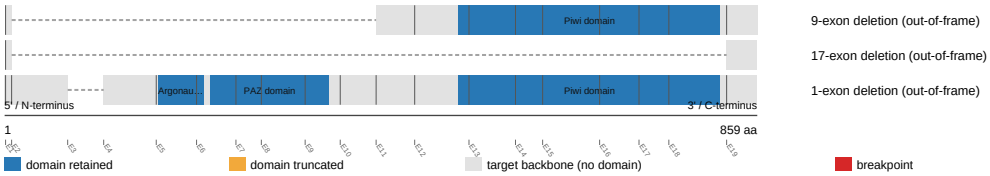


Partner-block colors are decorative; row labels identify partners.

intragenic deletion schematic

AGO2 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



9-exon deletion (out-of-frame)

17-exon deletion (out-of-frame)

1-exon deletion (out-of-frame)